

Chapter 14 – Multidimensional Unconstrained Optimization 多维无约束优化 Cheatsheet

◆ 14.1 Using the Gradient to Evaluate the Path of Steepest Ascent 利用梯度确定最速上升路径

公式:

1. 梯度 (Gradient):

$$\nabla f = \left(\frac{\partial f}{\partial x}\right)\mathbf{i} + \left(\frac{\partial f}{\partial y}\right)\mathbf{j}$$

2. 方向角度:

$$\tan \theta = \frac{\partial f / \partial y}{\partial f / \partial x} \Rightarrow \theta = \tan^{-1} \left(\frac{\partial f / \partial y}{\partial f / \partial x} \right)$$

单位换算: $1 \text{ 弧度} = \frac{180}{\pi} \approx 57.3^\circ$, $1^\circ = \frac{\pi}{180} \approx 0.01745 \text{ 弧度}$

3. 方向导数 (斜率):

$$g'(0) = \left(\frac{\partial f}{\partial x}\right) \cos \theta + \left(\frac{\partial f}{\partial y}\right) \sin \theta$$

4. 梯度模长:

$$|\nabla f| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Q14-1 示例:

- 函数: $f(x, y) = 2x^2 + y^2$
- 点: $(2, 2)$
- 方向向量: $\mathbf{h} = 3\mathbf{i} + 2\mathbf{j}$

步骤:

- 梯度: $\nabla f = 8\mathbf{i} + 4\mathbf{j}$
- 方向角: $\theta = \tan^{-1}(2/3) \approx 0.588 \text{ rad} = 33^\circ 41' 24''$
- $g'(0) = 8 \cos(0.588) + 4 \sin(0.588) = 8.8752$

Q14-2 示例:

- 函数: $f(x, y) = 2xy + 1.5y - 1.25x^2 - 2y^2 + 5$
- 点: $(0.8, 1.2)$
- 梯度: $\nabla f = 0.4\mathbf{i} - 1.7\mathbf{j}$
- $\theta = \tan^{-1}(-1.7/0.4) = -1.34 \text{ rad}$
- $g'(0) = 0.4 \cos(-1.34) - 1.7 \sin(-1.34) = 1.7464$
- 梯度模: $|\nabla f| = \sqrt{0.4^2 + (-1.7)^2} = 1.7464$

◆ 14.2 求偏导与解方程组

Q14-3 示例:

- $f(x, y) = 2.25xy + 1.75y - 1.5x^2 - 2y^2$

偏导数:

- $\frac{\partial f}{\partial x} = 2.25y - 3x = 0$
- $\frac{\partial f}{\partial y} = 2.25x + 1.75 - 4y = 0$

解线性方程组得:

- $y = \frac{4}{3}x \Rightarrow x = 0.5676, y = 0.7568$

◆ 14.3 沿梯度方向构建一维函数 $g(h)$

公式:

1. 梯度: $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$

2. 构建函数:

$$g(h) = f(x_0 + h\frac{\partial f}{\partial x}, y_0 + h\frac{\partial f}{\partial y})$$

3. 解 $g'(h) = 0$ 得最优步长 h^* , 再代入得到新点 (x_i, y_i)

Q14-11 示例:

- 函数: $f(x, y) = 6x^2y - 9y^2 - 8x^2$, 点: $(4, 2)$
- 梯度: $\nabla f = 32\mathbf{i} + 60\mathbf{j}$

构建函数:

- $x = 4 + 32h, y = 2 + 60h$
- $f(x, y) = 368640h^3 + 63856h^2 + 4624h + 28$

Q14-4 示例:

- 函数: $f(x, y) = 2.25xy + 1.75y - 1.5x^2 - 2y^2$, 起点: $(1, 2)$

步骤:

- 梯度: $\nabla f = 1.5\mathbf{i} - 4\mathbf{j}$
- 方向函数:

$$g(h) = -48.875h^2 + 18.25h - 1.5$$

- 解 $g'(h) = 0 \Rightarrow h^* = 0.186701$
- 新点: $(x_1, y_1) = (1 + 1.5h, 2 - 4h) = (1.280, 1.253)$

◆ 14.4 Optimal Steepest Ascent 最速上升法

公式:

- 梯度: $\nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$
- 一维函数: $g(h) = f(x_0 + h\partial f/\partial x, y_0 + h\partial f/\partial y)$
- Hessian判别:

$$|H| = (\frac{\partial^2 f}{\partial x^2})(\frac{\partial^2 f}{\partial y^2}) - (\frac{\partial^2 f}{\partial x \partial y})^2$$

- 极值判别:
 - $|H| > 0, \frac{\partial^2 f}{\partial x^2} < 0 \Rightarrow$ 极大值
 - $|H| > 0, \frac{\partial^2 f}{\partial x^2} > 0 \Rightarrow$ 极小值
 - $|H| < 0 \Rightarrow$ 鞍点

E14-3 示例:

- 函数: $f(x, y) = 2xy + 2x - x^2 - 2y^2$, 初始: $(-1, 1)$

步骤:

- 解方程组得极值点 $x = 2, y = 1$
- Hessian:
 - $\frac{\partial^2 f}{\partial x^2} = -2, \frac{\partial^2 f}{\partial y^2} = -4, \frac{\partial^2 f}{\partial x \partial y} = 2$
 - $|H| = 4 > 0, \frac{\partial^2 f}{\partial x^2} < 0 \Rightarrow$ 极大值

最速上升法迭代一:

- $\nabla f = (6, -6) \Rightarrow x = -1 + 6h, y = 1 - 6h$
- $g(h) = -180h^2 + 72h - 7 \Rightarrow h^* = 0.2$
- 新点: $x = 0.2, y = -0.2$

迭代二:

- $\nabla f = (1.2, 1.2) \Rightarrow x = 0.2 + 1.2h, y = -0.2 + 1.2h$
- $g(h) = -1.44h^2 + 2.88h + 0.2 \Rightarrow h^* = 1$
- 新点: $x = 1.4, y = 1$

◆ Q14-6 Steepest Descent Method 最速下降法

函数:

$$f(x, y) = (x-3)^2 + (y-2)^2$$

方法:

- 方向: 使用负梯度方向:

$$-\nabla f = -(\frac{\partial f}{\partial x})\mathbf{i} - (\frac{\partial f}{\partial y})\mathbf{j}$$

- 收敛判据:

$$\varepsilon_s = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100\% \quad \text{或} \quad \varepsilon_s = \left| \frac{y_{i+1} - y_i}{y_{i+1}} \right| \times 100\%$$